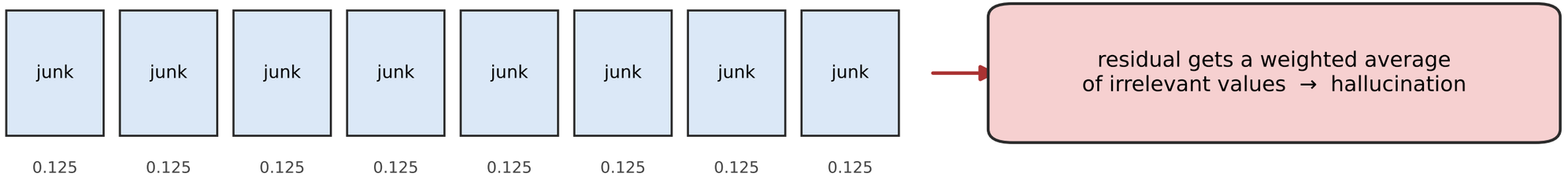
Softmax has no off switch. pn_fir_tau does.

Same query, no matching key: what each normalizer writes into the residual stream.

Softmax

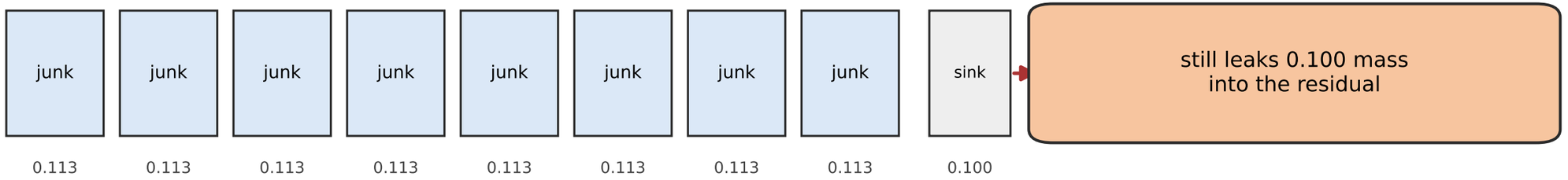
simplex: mass = 1.000



Hallucination rate on absent queries: 27.2–34.4% across simplex normalizers (Exp 5).

Virtual-sink Softmax

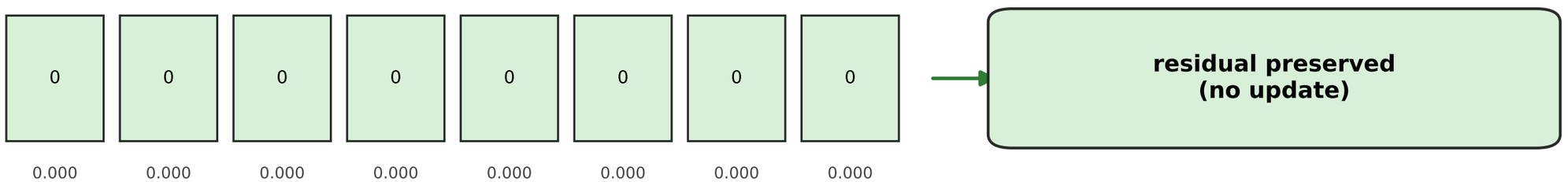
mass < 1, but tails stay positive



Residual leakage on absent keys: 0.100 ± 0.005 (Exp 5).

pn_fir_tau

sub-simplex: mass ≤ 1



Unanswerable-row mass: 0.002 ± 0.002; peak gate closes the row (Exp 5).

Controlled synthetic abstention task, 50% unanswerable queries; see the paper for scope.